

Computer Science 1

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2026-09-05

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Contact

- **Year:** 2026/27/1, Semester 1
- **Class Location:** P/106/A, Institute of Psychology, University of Debrecen
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1 Syllabus

Course Number: BTPS1210BA-E

Credits: 2

Course type: Practical

Year: 2026/27/1, Semester 1 (Autumn)

Class Location: P/106/A, Institute of Psychology, University of Debrecen

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1.1 Course overview

During the course, students will acquire the basic digital skills they will need for their studies. The course attempts to balance the knowledge levels of students with different backgrounds. Topics include file management, reproducible reporting with RStudio and Quarto, bibliography management, statistical data analysis, spreadsheets, word processing, and presentation design.

The main purpose of the course is to support statistics courses. Today, statistical analysis is carried out by computer, so knowledge of one or more statistical software packages is essential. Students will be introduced to the *jamovi* statistical software and to *R/RStudio*. The *jamovi* part of the course will cover data entry and preparation, descriptive analysis, and graphical analysis. In *RStudio*, students will learn to create Markdown-based documents and generate HTML output.

1.2 Attendance

Active participation is an important part of this class. Attendance is required. Students may miss no more than three practical sessions. If a student exceeds the maximum number of absences, a practical grade will not be given and a “refused” will be entered in the electronic learning system.

1.3 Assessment and Grades

During the semester there will be 4 homework assignments and one project. In the Table 1.1 you can see the details of the homework assignments and the project. The project will be presented during the last two classes of the semester. The Table 1.1 also shows the deadlines for the homework assignments and the project.

Table 1.1: Points and deadlines

Components	Points	Links	Dates
Homework 1	15	Details¹	October 14.
• RStudio reporting: markdown • Quarto document			
Homework 2	15	Details²	November 4.
• Jamovi (1) • JabRef			
Homework 3	15	Details³	November 25.
• Jamovi (2)			
Homework 4	15	Details⁴	December 7.
• Word • JabRef			
Project final artifact	30	Details⁵	November 9. and December 5.
Project final presentation	10	Details	Last two classes of the semester.
TOTAL	100		

There are many ways to earn **extra points**, which are added to the points shown in Table 1.1. Extra points can be earned in the following ways:

¹Hint: [My solution of Homework 1](#)

²Hint: [My solution of Homework 2](#)

³Hint: [My solution of Homework 3](#)

⁴Hint: [My solution of Homework 4](#)

⁵Hint: [My solution of Project final artifact.](#) [My solution of The answer.](#)

- active participation in class (up to 5 points)
- making and presenting a Kahoot! test (3 points each)
- winning a Kahoot! test (3 points each)
- sending a solution of a lab
 - Lab 01
 - * Problem 1: up to 1 point ([link](#))
 - * Problem 2: up to 1 point ([link](#))
 - * Problem 3: up to 2 points ([link](#))
 - Lab 02
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
 - Lab 03
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - Lab 04
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
 - Lab 05
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - Lab 06
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
 - Lab 07
 - * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))

- * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
 - * Problem 5: up to 3 points ([link](#))
 - * Problem 6: up to 3 points ([link](#))
 - * Problem 7: up to 3 points ([link](#))
 - * Problem 8: up to 3 points ([link](#))
- Lab 08
- * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
 - * Problem 5: up to 3 points ([link](#))
- Lab 09
- * Problem 1: up to 2 points ([link](#))
 - * Problem 2: up to 2 points ([link](#))
 - * Problem 3: up to 3 points ([link](#))
 - * Problem 4: up to 3 points ([link](#))
- Lab 10
- * Problem 1: up to 4 points ([link](#))
 - * Problem 2: up to 4 points ([link](#))
- Lab 11
- * Problem 1: up to 4 points ([link](#))
 - * Problem 2: up to 4 points ([link](#))

The Table [1.2](#) shows the grading scale. A student’s final grade will be determined by the total number of points earned during the semester. The final grade will be determined as follows:

Table 1.2: Grading

Points	Grade offered
0–50	1 (failed)
51–60	2
61–74	3
75–85	4
86–	5

Students can earn a maximum of 100 points through the homework assignments and the project work, to which extra points are added.

1.4 Course details

Table 1.3: Course details

Topic	Links
Intro	Details
Files, directories, compressing	Details
RStudio reporting: markdown, Quarto document	Details
JabRef - Bibliography management	Details
TXT, CSV, Spreadsheets	Details
Jamovi - Introduction	Details
Autumn break	
Jamovi - Import data, Data conversion	Details
Jamovi - Data transformation	Details
Jamovi - Descriptive statistics I.	Details
Jamovi - Descriptive statistics II.	Details
Word and PowerPoint	Word ; PPT
Project presentation	Details
Project presentation	Details

1.5 Textbooks

- **Required:**

- University of Helsinki (2023, p. 1 Introduction, Advanced studies: presenting data)
- Navarro & Foxcroft (2022, pp. 1–117)
- Poulson (2019, Getting started, Wrangling data, Exploration)

- **Optional texts:**

- Pajankar (2022)
- Cote et al. (2021, pp. 1–77)
- Gordon & Bataineh (2022, pp. 1–109)
- Landers (2024)

1.6 Software

The course requires some [software](#).

1.7 Prerequisites

No prerequisite is required. Prior knowledge in computer literacy (word processing, spreadsheets, presentation) is recommended but not required.

2 Course details

2.1 Intro

- Readings:
 - Navarro & Foxcroft (2022, pp. 3–12)

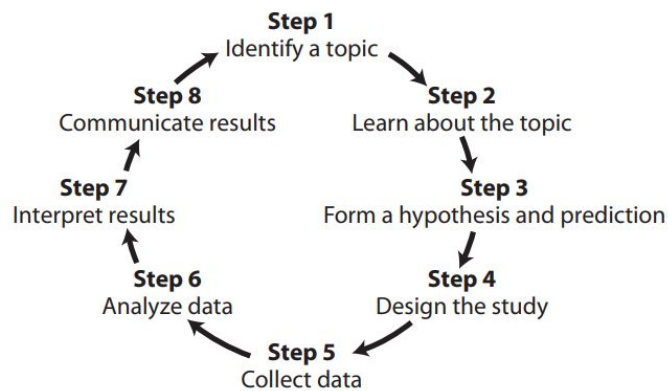


Figure 3.6 The steps in conducting scientific research

Figure 2.1: Schweigert (2021) pp. 58

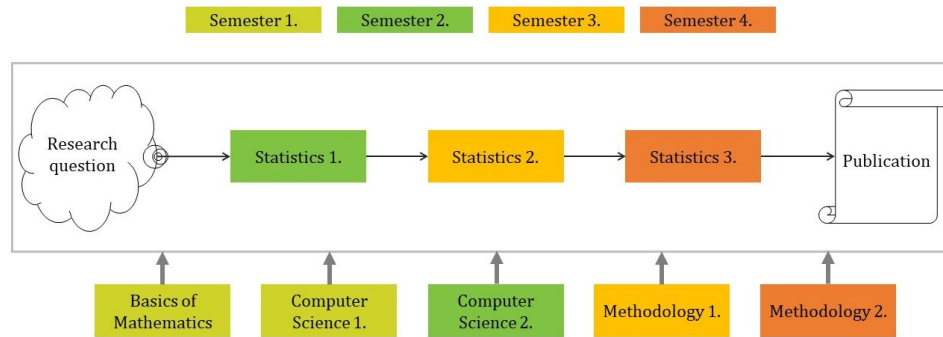


Figure 2.2: The course in the BA program

2.2 Files, directories, compressing

- Readings:
 - University of Helsinki (2023) [1.1 Computer functionality](#)
 - University of Helsinki (2023) [1.2. Using files and programs](#)
- Problems
 - [Lab 01](#)
- Questions
 - What is the purpose of buying a computer?
 - What problems does the computer solve?
 - How does a computer solve problems?
 - What is a file?
 - What is the purpose of the file?
 - What is a directory?
 - What is the purpose of the directory?
 - What is the structure of directories?
 - What are packing and unpacking (or compression and extraction)?
 - What is the purpose of compression?
- Skills
 - creating, renaming, copying, moving, and deleting files and directories
 - compressing and extracting files and directories
 - creating a file of a specific type (TXT, DOCX, XLSX, ODT, ODS, PDF, PPTX, PNG, JPG, OMV, BIB, etc.)

- typing special characters (ampersand, angle brackets, apostrophe, asterisk, at sign, backslash, braces, brackets, caret, dollar sign, em dash, en dash, hyphen, backtick, tilde, parentheses, percent, pipe, slash, underscore, etc.)

2.3 RStudio reporting

- Readings:
 - [RStudio for the Total Beginner \(only the first minute and a half\)](#)
 - [Posit Cheatsheets: RStudio IDE](#)
 - [The Markdown Guide: Getting Started, Markdown Cheat Sheet, Basic Syntax, Extended Syntax](#)
 - [Posit Cheatsheets: Publish and Share with Quarto](#)
 - [Tutorial: Hello, Quarto](#)
- Problems
 - [Lab 02](#)

2.4 JabRef - Bibliography management

- Readings:
 - [JabRef: Getting started](#)
 - [Quarto: Citations & Footnotes](#)
- Problems
 - [Lab 03](#)

2.5 TXT, CSV, Spreadsheets

- Readings:
 - [What is Tabular Data?](#)
 - University of Helsinki (2023) [P.3 Spreadsheets](#)
- Problems
 - [Lab 04](#)

2.6 Jamovi - Introduction

- Readings:
 - Navarro & Foxcroft (2022, pp. 43–49)
 - Poulson (2019)
 - * Welcome
 - * Installing jamovi
 - * Navigating jamovi
- Problems
 - [Lab 05](#)

2.7 Jamovi - Import data, Data conversion

- Readings:
 - Navarro & Foxcroft (2022, pp. 49–54)
 - Poulson (2019)
 - * Sample data
 - * jamovi modules
 - * Entering data
 - * Importing data
 - * Variable types & labels
- Problems
 - [Lab 06](#)

2.8 Jamovi - Data transformation

- Readings:
 - Navarro & Foxcroft (2022, pp. 97–117)
 - Poulson (2019)
 - * Wrangling data: chapter overview
 - * Computing means
 - * Computing z-scores
 - * Transforming scores to categories
 - * Filtering cases
- Problems
 - [Lab 07](#)

2.9 Jamovi - Descriptive statistics I.

- Readings:
 - Navarro & Foxcroft (2022, pp. 59–84)
 - Poulson (2019)
 - * Exploration: chapter overview
 - * Descriptive statistics
- Problems
 - [Lab 08](#)

2.10 Jamovi - Descriptive statistics II.

- Readings:
 - Navarro & Foxcroft (2022, pp. 85–96)
 - Poulson (2019)
 - * Histograms
 - * Density plots
 - * Box plots
 - * Violin plots
 - * Dot plots
 - * Bar plots
 - * Exporting tables & plots
- Problems
 - [Lab 09](#)

2.11 Word processing - Styles, TOC, Bibliography, APA7

- Readings:
 - University of Helsinki (2023) [P.2 Word processing](#)
 - [Study skills: APA7](#)
 - [Citation: About APA 7th ed.](#)
- Problems
 - [Lab 10](#)

2.12 Presentation: PowerPoint

- Readings:
 - University of Helsinki (2023) [P.4 Slide shows](#)
- Problems
 - [Lab 11](#)

2.13 Project presentation

3 Software

Please install the following software on your computer:

- **Jamovi** [Available from url: <https://www.jamovi.org/download.html>. Choose the *current* version.]
- **R** [Available from url: <https://cloud.r-project.org/>.]
- **RStudio** [Available from url: <https://posit.co/downloads/>. Choose the free *RStudio Desktop*.]
- **JabRef** [Available from url: <https://www.jabref.org/#download>.]
- **Zotero** [Available from url: <https://www.zotero.org/downloads>. *Zotero* is an alternative to *JabRef*.]
- **Mendeley Desktop** [Available from url: <https://www.mendeley.com/download-reference-manager>. *Mendeley Desktop* is an alternative to *JabRef*. We won't use it in the course, but it's a good reference manager.]

4 Schedule of the Academic Year

The academic year in Hungary starts in September and is built up of two semesters: autumn and spring semesters. Both semesters start with a registration period which is followed by the study period and an examination period afterwards. [Details in Hungarian.](#)

I. (AUTUMN) SEMESTER

- Free course registration and withdrawal period: 31 August, 2026 (Monday), 8:00 – 20 September, 2026 (Sunday), 23:59 (3 weeks)
 - Non-graduating students may register for courses from 13:00 on 1 September, 2026.
- Registration week: 31 August – 4 September, 2026 (1 week)
- Part-time education begins on 3 September, 2026 (Thursday).
- Study period:
 - Non-graduating students: 7 September – 11 December, 2026 (13 weeks)
 - Graduating students completing their studies in January 2027: 7 September – 27 November, 2026 (11 weeks)
- Autumn break: 19 – 22 October, 2026 (1 week)
- Examination period:
 - Non-graduating students: 14 December, 2026 – 29 January, 2027 (7 weeks)
 - Final examination period for graduating students completing their studies in January 2027: 28 December, 2026 – 15 January, 2027 (3 weeks)

II. (SPRING) SEMESTER

- Registration week: 1 – 5 February, 2027 (1 week)
- Study period for non-graduating students: 8 February – 14 May, 2027 (13 weeks)
- Spring break: 30 March – 2 April, 2027 (1 week)
- Examination period for non-graduating students: 18 May – 2 July, 2027 (7 weeks)

5 Learning with AI: Gemini Notebooks and NotebookLM

5.1 What is NotebookLM?

[Google NotebookLM](#) is an AI-powered research and learning assistant built with Gemini models. A notebook is a workspace containing the sources you choose. You can ask questions about those sources and turn them into study materials such as summaries, study guides, mind maps, flashcards, quizzes, and Audio Overviews.

NotebookLM notebooks may also be available from Gemini, depending on your account and region. There is an important difference between the two ways of using them: NotebookLM answers are grounded in the selected notebook sources, while Gemini may also use web search and other tools. This chapter therefore focuses on NotebookLM when the goal is to study the material in this course. See Google's explanation of the [NotebookLM–Gemini integration](#).

Source grounding makes an answer easier to check, but it does not make the answer automatically correct. NotebookLM can misunderstand a source, overlook a detail, or produce an inaccurate explanation. Use it as a study partner, not as an authority.

! The central rule

Do not ask AI to replace learning. Ask it to help you **retrieve, explain, compare, practise, and verify** what you are learning. Always open and read the cited part of the original course material before relying on an answer.

5.2 What you need

To use NotebookLM for this course, you need:

- a personal Google Account or a university-managed Google Account;
- access to NotebookLM enabled for that account;
- a supported web browser and an internet connection;
- the course materials you want to study; and
- enough time to verify AI-generated answers against the original sources.

If you use a university-managed account and cannot open NotebookLM, access may need to be enabled by the institution’s administrator. Availability and individual features can also depend on age, account type, region, and subscription. Check the current requirements in the [official NotebookLM help](#).

NotebookLM accepts sources including PDF, Markdown, Word, PowerPoint, CSV, images, pasted text, Google Docs, Google Slides, Google Sheets, web pages, public YouTube videos with captions, and audio files. The supported types and limits may change; the current list is available in the guide to [adding sources](#).



Privacy and copyright

Upload only material you are allowed to use. Do not upload passwords, grades, Neptun identifiers, private correspondence, personal data about classmates, confidential research data, or copyrighted material that you are not permitted to share. Before using an institutional account, follow the university’s data-protection and acceptable-use rules.

5.3 Create a course notebook

5.3.1 1. Choose a useful scope

Start with one notebook named, for example, `Computer Science 1 -- 2026/27/1`. A single notebook is convenient for finding connections across the course. If it becomes difficult to navigate, create smaller notebooks for major topics, such as:

- Files, RStudio, and Quarto;
- Data and jamovi;
- References and academic documents; and
- Spreadsheets, Word, and presentations.

Smaller notebooks make it easier to select the relevant sources and reduce the chance that unrelated material influences an answer.

5.3.2 2. Add the main book

The simplest starting point is the [PDF version of this book](#). Download it and add it as a PDF source. It contains the course overview, syllabus, software information, academic calendar, and the main book chapters.

The laboratory and assignment pages are separate HTML documents and are not all contained in the book PDF. Add the pages needed for the topic you are studying. You can open a page

from the syllabus or course schedule, copy its web address, and add that URL as a source. For example, a notebook about data transformation could include:

- [Lab 06: Exploring Data](#);
- [Lab 07: Transforming Data](#); and
- [Homework 3](#).

When a web page is added by URL, NotebookLM normally imports the text of that page, not every page linked from it. Images and embedded content may also be omitted. Add each important page separately, or save it as a PDF and upload the PDF when visual details are essential.

5.3.3 3. Name and organise the sources

Give each source a short, descriptive name, such as **Syllabus**, **Lab 07 -- Transforming Data**, or **Homework 3**. Before asking a question, select only the sources relevant to it. Mention source names in the prompt when you want a focused answer.

If the course book or a laboratory page changes, update the source. A web page or uploaded local file is generally a snapshot of its content at the time it was added. Google Drive sources can offer synchronisation, but you should still verify that NotebookLM is using the latest version.

5.4 A reliable learning workflow

Use the following cycle for each topic:

1. **Preview.** Ask for a short map of the topic: key ideas, tools, prerequisites, and the order in which to learn them.
2. **Read.** Study the relevant section of the book or lab yourself.
3. **Retrieve.** Close the source and explain the topic from memory, or answer questions without assistance.
4. **Check.** Ask NotebookLM to compare your explanation with the selected sources and identify omissions or misconceptions.
5. **Apply.** Complete a new task in RStudio, jamovi, Word, a spreadsheet, or PowerPoint without copying an AI-generated final answer.
6. **Verify.** Follow NotebookLM's citations back to the course material and confirm every important claim or procedural step.
7. **Reflect.** Record what you can now do independently and what still needs practice.

This cycle is more effective than repeatedly requesting summaries because it requires active recall and practical use of the material.

5.5 How to write useful prompts

A good prompt specifies five things:

1. **Role:** what kind of help you want;
2. **Sources:** which selected materials should be used;
3. **Task:** what NotebookLM should do;
4. **Output:** the desired structure and level; and
5. **Verification:** how claims should be linked to the sources.

For example:

Act as a patient tutor for a first-year psychology student. Use only Lab 07 -- **Transforming Data**. Explain computed variables in plain language, give one new example, and end with three questions that test understanding. Cite the source for each procedural claim.

Specific prompts usually produce more useful results than requests such as “Explain this” or “Summarise everything.”

5.6 Examples for this course

5.6.1 Understand a new topic

Using Lab 01, explain the difference between a file, a directory, and a file extension. First use an everyday analogy, then give a technically accurate explanation. Finish with five short self-test questions. Do not show the answers until I ask for them.

5.6.2 Learn RStudio and Quarto

Use the RStudio and Quarto course materials. Create a checklist for starting a new Quarto project and rendering an HTML document. For each step, explain why it is necessary and cite the relevant passage.

If a render fails, do not ask NotebookLM merely to “fix it.” Paste the exact error message after removing personal paths or data, and ask:

Based only on the selected course sources, list the three most likely causes of this error. For each cause, give one safe diagnostic check. Separate statements supported by the sources from your own inference.

5.6.3 Practise data concepts and jamovi

Use Labs 05–07. Give me eight scenarios in which I must identify the variable type or choose an appropriate data transformation. Ask one question at a time. After each answer, tell me what was correct, what needs improvement, and where the relevant rule appears in the sources.

Compare renaming a variable, recoding levels, computing a new variable, transforming scores into categories, and filtering cases. Present the result as a table with purpose, input, output, and one example for each.

5.6.4 Prepare for a homework assignment

Use Homework 3 and the relevant jamovi labs. Turn the assignment into a sequence of checkpoints. Do not solve the assignment or choose a dataset for me. At each checkpoint, ask me to explain my decision before you give feedback.

This keeps ownership of the work with the student while still providing structure and formative feedback.

5.6.5 Improve bibliography management

Use the bibliography-management materials. Explain the workflow from a DOI to a checked bibliography entry and an in-text citation. Create a list of common metadata errors I should check manually in JabRef or Zotero.

5.6.6 Prepare a presentation

Use the presentation lab and the project instructions. Create a review rubric for my slides covering structure, readability, visual consistency, evidence, and timing. Ask me questions that help me evaluate my own work; do not write the presentation for me.

5.6.7 Plan study time

Use the syllabus and academic calendar. Make a two-week study plan for the next assignment. Include reading, active-recall questions, software practice, and a final verification step. Clearly distinguish official deadlines from the study dates you recommend.

Always confirm deadlines directly in the current [syllabus](#) and [academic calendar](#). Do not rely on an older NotebookLM source or chat response for date-sensitive information.

5.7 Study materials in the Studio panel

NotebookLM can transform selected sources into several kinds of learning materials. Availability may vary by account.

Tool	A productive use in this course	What to check
Study guide or briefing	Create an overview before reading a lab	Does it include the practical steps, not only definitions?
Mind map	See relationships among files, projects, data, analyses, and outputs	Are the relationships actually supported by the sources?
Flashcards	Practise terminology, menu locations, and concepts	Can you answer before turning over the card?
Quiz	Test understanding after studying	Does the explanation justify the correct answer?
Audio Overview	Review concepts while travelling or walking	Recheck technical instructions visually before applying them

An Audio Overview can be customised by topic, level, language, and format. For example:

Create a concise Audio Overview for a beginner. Focus on the difference between data preparation and descriptive analysis in jamovi. Explain why each step matters, but do not invent menu commands not present in the selected labs.

Audio is useful for review, but it is not a substitute for operating the software. AI-generated audio can contain inaccuracies or glitches; Google describes the available options and limitations in the [Audio Overview guide](#).

5.8 How to evaluate an AI answer

Before accepting an answer, ask:

- Did the answer use the correct and current course sources?
- Does every important claim have a citation?
- Does the cited passage actually support the claim?
- Can I reproduce the procedure in the relevant software?

- Did the answer confuse an example with a general rule?
- Did it invent a menu item, deadline, filename, or submission method?
- Can I explain the answer in my own words without looking at it?

Useful follow-up prompts include:

Quote the shortest relevant passage for each step and explain how it supports your answer.

Which parts of your previous answer are direct statements from the sources, and which parts are inferences?

Is this question fully answered by the selected sources? If not, say what information is missing instead of guessing.

If the cited source and the answer disagree, trust the current course source and inform the instructor if the source itself appears inconsistent.

5.9 Academic integrity

NotebookLM may help you understand instructions, practise skills, generate self-test questions, review your reasoning, and find relevant passages. It should not complete assessed work on your behalf.

For homework and the project:

- choose and inspect your own article or dataset;
- make your own analytical and design decisions;
- perform the work yourself in the required software;
- verify every result against the data and course requirements;
- write the submitted explanation in your own words; and
- follow any AI-use rules given by the instructor.

Never submit AI-generated content that you do not understand or cannot reproduce. The aim of this course is not merely to produce files, but to develop skills you can use independently.

5.10 A 30-minute study session

Here is a simple routine you can reuse:

1. **5 minutes:** select one topic and the relevant sources.
2. **5 minutes:** ask for a concept map or three guiding questions.
3. **10 minutes:** study the original material and practise in the software.

4. **5 minutes:** answer a quiz or explain the procedure from memory.
5. **5 minutes:** check citations, correct mistakes, and write down the next step.

End the session with one sentence beginning: “I can now independently...” If you cannot complete that sentence honestly, the next session should include another practical attempt rather than another summary.

5.11 Final perspective

The most valuable feature of a Gemini-powered notebook is not that it can produce a quick answer. Its value is that it can turn the course book into an interactive learning environment: it can ask questions, connect ideas, adapt explanations, and guide you back to the relevant source. Used well, it shortens the distance between reading and practice while keeping the student responsible for verification, judgement, and the final work.

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